

# FUNDAMENTALS OF WEB DESIGN (415)

## STATE 2024

|                                      |                           |
|--------------------------------------|---------------------------|
| Multiple Choice (25 @ 2 points each) | _____ (50 points)         |
| Application                          | _____ (250 points)        |
| <b>TOTAL POINTS</b>                  | <b>_____ (300 points)</b> |

**Test Time: 90 minutes**

**GENERAL GUIDELINES:**

*Failure to adhere to any of the following rules will result in disqualification:*

1. Member must hand in this test booklet and all printouts if any. Failure to do so will result in disqualification.
2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests (handwritten, photocopied, or keyed) are allowed in the testing area.
3. Electronic devices will be monitored according to ACT standards.

**MULTIPLE CHOICE ANSWERS**

1. A
2. C
3. C
4. B
5. B
6. A
7. B
8. A
9. A
10. A
11. C
12. A
13. A
14. A
15. B
16. B
17. C
18. A
19. B
20. A
21. D
22. B
23. B
24. A
25. A

## APPLICATION

### Overview

You are tasked with creating a modern, responsive web page titled "Explore Chicago." This page should showcase various attractions, historical sites, and fun facts about Chicago, using advanced HTML5 and CSS3 techniques to create a visually appealing, accessible, and informative experience.

### Design Requirements

1. Modern Layout:
  - Employ Flexbox for layout design in various sections.
  - Include a **header**, **main**, and **footer**.
2. Navigation Bar:
  - Located in the **header**.
  - Create jump links for the following: **Attractions**, **History**, **Fun Facts**, **Contact**
3. Showcase Area:
  - A banner section with a background image ("images/chicago-banner.jpg").
  - Heading: "Welcome to Chicago" and subheading: "Explore the city of winds".
4. Content Sections:
  - Located in the **main**.
  - Three sections in **main**: **Attractions**, **History**, **Fun Facts**
  - **Attractions**
    - a. Title
      - i. Discover the Wonders of Chicago
    - b. Text Content
      - i. Willis Tower: An iconic skyscraper offering stunning skyline views. A must-visit for breathtaking sights.
      - ii. The Art Institute of Chicago: Home to an impressive collection of artworks, spanning thousands of years.
      - iii. Navy Pier: A vibrant location filled with amusement rides, shops, and eateries, perfect for family fun.
    - c. Image
      - i. chicago-attractions.jpg
    - d. Table Content
      - i. Landmark, Opening Hours, Ticket Price
      - ii. Willis Tower, 9 AM - 10 PM, \$25
      - iii. The Art Institute of Chicago, 10 AM - 5 PM, \$20
      - iv. Navy Pier, 10 AM - 8 PM, Free Entry
  - **History**
    - a. Title
      - i. A Journey Through Time in Chicago
    - b. Text Content

- i. From its founding in 1833 to becoming a bustling metropolis, Chicago's history is as rich as it is diverse.
    - ii. Great Chicago Fire (1871): A major event that led to the extensive rebuilding of the city.
    - iii. World's Columbian Exposition (1893): A landmark world fair that showcased Chicago's resurgence.
  - c. Image
    - i. chicago-history.jpg
- **Fun Facts**
  - a. Title
    - i. Fun Facts About Chicago
  - b. Text Content
    - i. Chicago's nickname "The Windy City" was coined not just for its weather, but also for its long-winded politicians.
    - ii. The first Ferris wheel made its debut in Chicago in 1893.
    - iii. Chicago River is the only river in the world that flows backward, a result of an engineering marvel.
  - c. Image
    - i. chicago-funfacts.jpg

5. Styling:

- External stylesheet.
- Consistent color scheme and typography.

6. Images:

- All images must have alt text with a basic photo description.
- Images should be stored and called from the **images** folder.

7. Footer:

- Contact information
  - a. Contact Information: Chicago Tourism Office, 123 City St, Chicago, IL, 60601, (312) 555-1234, info@chicagotourism.com

8. Accessibility and Responsive Design:

- Ensure website accessibility (text alternatives for images, keyboard navigability).
- Use media queries for responsiveness on different devices.

| Criteria                    | Description                                                                                                                                  | Points Possible | Points Earned |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|-----------------|---------------|
| <b>Layout and Structure</b> | Modern layout using Flexbox, correct implementation of header, main, footer.                                                                 | 40              |               |
| <b>Responsiveness</b>       | Effective use of media queries, consistent design on various screen sizes.                                                                   | 30              |               |
| <b>Navigation Bar</b>       | Correct placement and linking, CSS transitions for hover effects.                                                                            | 20              |               |
| <b>Showcase Area</b>        | Inclusion and styling of banner with image and text, creativity, and visual appeal.                                                          | 30              |               |
| <b>Content Sections</b>     | Accurate replication of provided content, styling, and layout of sections.                                                                   | 50              |               |
| <b>Styling</b>              | Use of modern design elements, consistency in color scheme and typography.                                                                   | 30              |               |
| <b>Images and Tables</b>    | Correct implementation and alt text for images, effective use of a table in Attractions section with specific landmarks, hours, and pricing. | 25              |               |
| <b>Footer</b>               | Inclusion of contact information.                                                                                                            | 15              |               |
| <b>Accessibility</b>        | Adherence to accessibility standards.                                                                                                        | 20              |               |
|                             | <b>Total</b>                                                                                                                                 | <b>250</b>      |               |

ATTRACTIONS   HISTORY   FUN FACTS   CONTACT



# Welcome to Chicago

## Explore the city of winds

### Discover the Wonders of Chicago

Willis Tower: An iconic skyscraper offering stunning skyline views. A must-visit for breathtaking sights.

The Art Institute of Chicago: Home to an impressive collection of artworks, spanning thousands of years.

Navy Pier: A vibrant location filled with amusement rides, shops, and eateries, perfect for family fun.



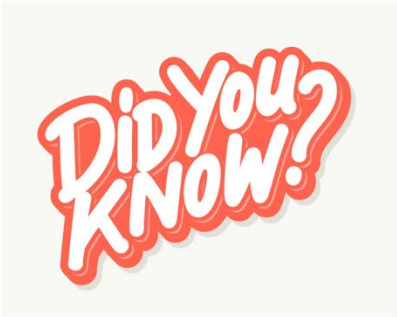
| Landmark                     | Opening Hours | Ticket Price |
|------------------------------|---------------|--------------|
| Willis Tower                 | 9 AM - 10 PM  | \$25         |
| The Art Institute of Chicago | 10 AM - 5 PM  | \$20         |
| Navy Pier                    | 10 AM - 8 PM  | Free Entry   |

### A Journey Through Time in Chicago

- From its founding in 1833 to becoming a bustling metropolis, Chicago's history is as rich as it is diverse.
- Great Chicago Fire (1871): A major event that led to the extensive rebuilding of the city.
- World's Columbian Exposition (1893): A landmark world fair that showcased Chicago's resurgence.



### Fun Facts About Chicago



- Chicago's nickname "The Windy City" was coined not just for its weather, but also for its long-winded politicians.
- The first Ferris wheel made its debut in Chicago in 1893.
- Chicago River is the only river in the world that flows backward, a result of an engineering marvel.

Contact Information: Chicago Tourism Office, 123 City St, Chicago, IL, 60601, (312) 555-1234, info@chicagotourism.com

SAMPLE SOURCE CODE: COMPETITOR CODE WILL VARY

index.html

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
  <meta charset="UTF-8">
```

```
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
  <title>Explore Chicago</title>
```

```
  <link rel="stylesheet" href="style.css">
```

```
</head>
```

```
<body>
```

```
  <header>
```

```
    <nav>
```

```
      <a href="#attractions">Attractions</a>
```

```
      <a href="#history">History</a>
```

```
      <a href="#fun-facts">Fun Facts</a>
```

```
      <a href="#contact">Contact</a>
```

```
    </nav>
```

```
  </header>
```

```
  <div class="showcase">
```

```
    <h1>Welcome to Chicago</h1>
```

```
    <p>Explore the city of winds</p>
```

```
  </div>
```

```
  <div class="container">
```

```
    <main>
```

```
      <section id="attractions">
```

```
        <h2>Discover the Wonders of Chicago</h2>
```

```
        <p>Willis Tower: An iconic skyscraper offering stunning skyline views. A must-visit for  
breathhtaking sights.</p>
```

```
        <p>The Art Institute of Chicago: Home to an impressive collection of artworks, spanning  
thousands of years.</p>
```

<p>Navy Pier: A vibrant location filled with amusement rides, shops, and eateries, perfect for family fun.</p>



<table>

<tr>

<th>Landmark</th>

<th>Opening Hours</th>

<th>Ticket Price</th>

</tr>

<tr>

<td>Willis Tower</td>

<td>9 AM - 10 PM</td>

<td>\$25</td>

</tr>

<tr>

<td>The Art Institute of Chicago</td>

<td>10 AM - 5 PM</td>

<td>\$20</td>

</tr>

<tr>

<td>Navy Pier</td>

<td>10 AM - 8 PM</td>

<td>Free Entry</td>

</tr>

</table>

</section>

<section id="history">

<h2>A Journey Through Time in Chicago</h2>

<div class="history-content">

<div class="history-text">

<ul>

<li>From its founding in 1833 to becoming a bustling metropolis, Chicago's history is as rich as it is diverse.</li>

<li>Great Chicago Fire (1871): A major event that led to the extensive rebuilding of the city.</li>

<li>World's Columbian Exposition (1893): A landmark world fair that showcased Chicago's resurgence.</li>

</ul>

</div>

<div class="history-image">



</div>

</div>

</section>

<section id="fun-facts">

<h2>Fun Facts About Chicago</h2>

<div class="funfacts-content">

<div class="funfacts-image">



</div>

<div class="funfacts-text">

<ul>

<li>Chicago's nickname "The Windy City" was coined not just for its weather, but also for its long-winded politicians.</li>

<li>The first Ferris wheel made its debut in Chicago in 1893.</li>

<li>Chicago River is the only river in the world that flows backward, a result of an engineering marvel.</li>

</ul>

</div>

</div>

</section>

</main>

</div>

```
<footer id="contact">

    <p>Contact Information: Chicago Tourism Office, 123 City St, Chicago, IL, 60601, (312) 555-1234,
    info@chicagotourism.com</p>

</footer>

</body>

</html>
```

style.css

```
/* Basic Reset */

* {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
}

body {
    font-family: 'Arial', sans-serif;
    line-height: 1.6;
    color: #333;
    background-color: #f4f4f4;
}

/* Container */

.container {
    max-width: 1100px;
    margin: auto;
    overflow: hidden;
    padding: 0 20px;
}

/* Header */

header {
    background: #35424a;
    color: #ffffff;
```

```
padding: 20px 0;

display: flex;

justify-content: center;

align-items: center;

}

header nav {

display: flex;

align-items: center;

justify-content: center;

}

header a {

color: #ffffff;

text-decoration: none;

text-transform: uppercase;

font-size: 16px;

margin: 0 15px; /* Adjust spacing between links */

}

header a:hover {

color: #e8491d;

transition: color 0.3s ease-in;

}

/* Showcase */

.showcase {

background: url('images/chicago-banner.jpg') no-repeat center center/cover;

min-height: 300px;

text-align: center;

color: #ffffff;

padding-top: 50px;

}

.showcase h1 {
```

```
        font-size: 75px;
        margin-bottom: 10px;
    }
    .showcase p {
        font-size: 50px;
    }
    /* Main Content */
    main {
        padding: 20px;
        display: flex;
        flex-direction: column;
        gap: 20px;
    }
    section {
        background: #ffffff;
        padding: 20px;
        border-radius: 10px;
        box-shadow: 0px 0px 10px 0px #aaaaaa;
    }
    /* History Section */
    .history-content {
        display: flex;
        align-items: center;
        justify-content: space-between;
        gap: 20px;
    }
    .history-text {
        flex: 1;
    }
    .history-image {
```

```
        flex: 1;

        text-align: right;
    }

.history-image img {
    max-width: 100%;
    height: auto;
    border-radius: 5px;
}

/* Funfacts Section */

.funfacts-content {
    display: flex;
    align-items: center;
    justify-content: space-between;
    gap: 20px;
}

.funfacts-text {
    flex: 1;
}

.funfacts-image {
    flex: 1;
    text-align: left;
}

.funfacts-image img {
    max-width: 100%;
    height: auto;
    border-radius: 5px;
}

/* Footer */

footer {
    background: #35424a;
```

```
        color: #ffffff;

        text-align: center;

        padding: 20px 0;

        display: flex;

        flex-direction: column;

        align-items: center;

        justify-content: center;
    }

    footer p, footer a {

        margin-bottom: 10px;

    }

    footer a {

        color: #e8491d;

        text-decoration: none;

    }

    footer a:hover {

        color: #ffffff;

        text-decoration: underline;

    }

    /* Table */

    table {

        width: 100%;

        border-collapse: collapse;

        margin-top: 20px;

    }

    table, th, td {

        border: 1px solid #dddddd;

    }

    th, td {

        padding: 10px;
```

```
        text-align: left;
    }
    /* Images */
    img {
        max-width: 100%;
        height: auto
```